UC SAN DIEGO / ECE 251C

1

Filters and LTI systems

Executive summary

Linearity and time invariance describe a filter through its impulse response: the output is the convolution of that response with the input. Complex exponentials are eigenfunctions of this operation, so the Fourier transform turns filtering into multiplication by a frequency response. A two-sample moving average illustrates both descriptions. We then characterize linear phase for real FIR filters by pairing coefficients about a center of symmetry. Symmetry or antisymmetry, together with odd or even length, gives the four familiar types and explains the zeros that some types must have.

1.1 Signals, systems, and linearity

A discrete-domain signal is a sequence $x = (x[n])_{n\in\mathbb{Z}}$, where $\mathbb{Z} = \{\ldots, -1, 0, 1, \ldots\}$. We distinguish the sequence x from an individual sample $x[n]$. When n indexes time, we also call x a discrete-time signal. A system H acts on the sequence and produces another sequence $y = \mathrm{H}x$. Thus, the value of an output sample can depend on more than one input sample.

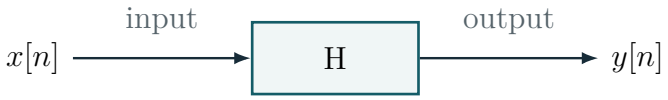


Figure 1.1: The system acts on the whole sequence: $y = \mathrm{H}x$.

Linearity. If $y_1 = \mathrm{H}x_1$ and $y_2 = \mathrm{H}x_2$, then linearity means

$$\mathrm{H}(\alpha x_1 + \beta x_2) = \alpha y_1 + \beta y_2 \qquad (\alpha, \beta \in \mathbb{R}). \tag{1.1}$$

In other words, the system commutes with linear combinations: we may combine the inputs before applying H, or combine their outputs afterward. In particular, choosing both coefficients to be zero shows that a linear system sends the zero sequence to the zero sequence.

1.2 Time invariance and the impulse response

Linearity describes the response to combinations of signals. Time invariance describes the response to translations of a signal. The latter property means that shifting an input changes only the location of the output, not its shape or amplitude.

For an integer k, define the shift T_k by

$$(T_k x)[n] \coloneqq x[n-k].$$

When $k > 0$, this delays the signal by k samples. If $y = \mathrm{H}x$, time invariance requires $\mathrm{H}(T_k x) = T_k y$. Equivalently,

$$\mathrm{H}T_k = T_k \mathrm{H} \qquad (k \in \mathbb{Z}). \tag{1.2}$$

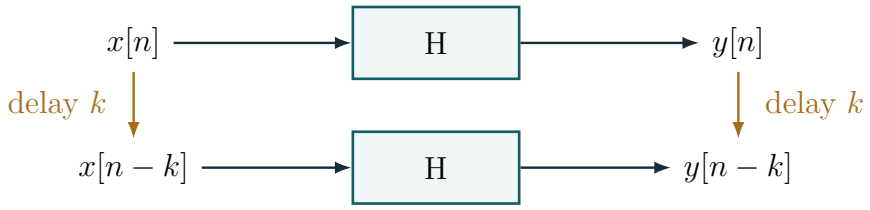


Figure 1.2: Delaying the input delays the output by the same amount. The two routes from the upper-left signal to the lower-right signal agree.

We call a system *linear time invariant*, or *LTI*, when it has both properties. The informal statement that it "does not care when it is turned on" means precisely the shift identity (1.2); we are comparing shifted whole input sequences, not changing an internal initial state.

Impulse response. The unit impulse, or Kronecker delta, is the sequence

$$\delta[n] \coloneqq \begin{cases} 1, & n = 0, \\ 0, & n \neq 0. \end{cases}$$

The *impulse response* is the output $h \coloneqq \mathrm{H}\delta$ when the input is δ.

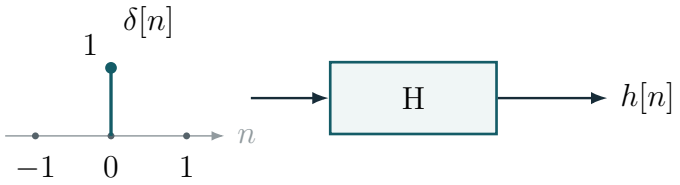


Figure 1.3: One unit sample at time zero probes the system. Its output is the impulse response.

Once h is known, time invariance determines the response to $\delta[n-k]$: it is $h[n-k]$. Linearity then determines the response to $a\delta[n-k]$: it is $ah[n-k]$. This observation is useful because an arbitrary finitely supported signal is a linear combination of shifted impulses.

1.3 Why an LTI system is described by convolution

Every finitely supported sequence can be assembled from impulses:

$$x[n] = \sum_{k\in\mathbb{Z}} x[k]\delta[n-k]. \tag{1.3}$$

For a fixed output index n, the factor $\delta[n-k]$ is zero unless $k=n$, in which case it equals one. Hence the sum returns $x[n]$. Although the formula is indexed by all integers, only finitely many coefficients $x[k]$ are nonzero under the present assumption.

Apply the system to this finite linear combination. First use linearity, and then time invariance:

$$\begin{aligned} y[n] &= \left[\mathrm{H}\left(\sum_k x[k]T_k\delta\right)\right][n] \\ &= \sum_k x[k](\mathrm{H}T_k\delta)[n] \\ &= \sum_k x[k](T_k\mathrm{H}\delta)[n] \\ &= \sum_k x[k]h[n-k]. \end{aligned} \tag{1.4}$$

Changing the summation index to $r=n-k$ gives

$$\boxed{y[n] = (h*x)[n] \coloneqq \sum_{r\in\mathbb{Z}} h[r]x[n-r].} \tag{1.5}$$

The operation in (1.5) is called *convolution*. It expresses the output entirely in terms of the input samples and the impulse response. In particular, the same coefficients $h[r]$ are used at every output index n; only the locations of the input samples change.

Infinite-duration inputs. The derivation above uses a finite sum of input impulses. Extending that derivation to infinite-duration inputs requires the system to respect the relevant limit of finite sums. For an FIR filter, the right side of (1.5) has finitely many terms and defines the convolution system for every

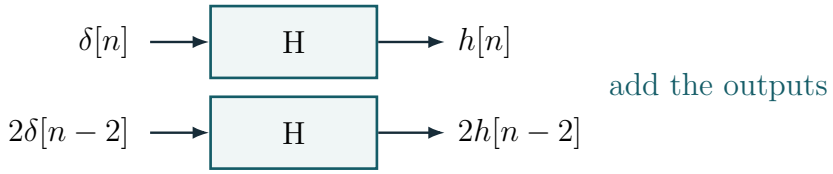


Figure 1.4: For $x[n] = \delta[n] + 2\delta[n-2]$, the output is $y[n] = h[n] + 2h[n-2]$. The same construction works for any finite collection of impulses.

input sequence. For infinite filters, a sufficient condition for pointwise absolute convergence is $\sum_r |h[r]| < \infty$ and bounded x, since

$$\sum_r |h[r]x[n-r]| \le \Big(\sup_m |x[m]|\Big) \sum_r |h[r]| < \infty.$$

1.4 FIR, IIR, and the moving average

The convolution formula tells us which input samples contribute to an output sample: they are the samples multiplied by nonzero values of h. The impulse response h is also called the *filter*. Its *support* is the set of indices where its coefficients are nonzero:

$$\operatorname{supp}(h) := \{n \in \mathbb{Z} : h[n] \neq 0\}.$$

A filter is *finite impulse response* (FIR) if $|\operatorname{supp}(h)| < \infty$, and *infinite impulse response* (IIR) if $|\operatorname{supp}(h)| = \infty$. Here $|\operatorname{supp}(h)|$ counts the elements of the support; it does not sum the coefficient magnitudes.

The distinction concerns the response to *one impulse*, not the duration of the input. An FIR filter may process an input that lasts forever, but every output sample depends on only finitely many input samples.

Example: a two-sample moving average. Consider

$$y[n] = \frac{1}{2}\big(x[n] + x[n-1]\big). \tag{1.6}$$

Each output is the average of the current input sample and the preceding one. To identify the filter, we need only evaluate the system on the unit impulse. Substituting $x = \delta$ gives

$$h[n] = \frac{1}{2}\delta[n] + \frac{1}{2}\delta[n-1], \qquad \operatorname{supp}(h) = \{0, 1\}.$$

Thus it is a two-tap FIR filter. Conversely, substituting these two nonzero coefficients into (1.5) gives exactly (1.6).

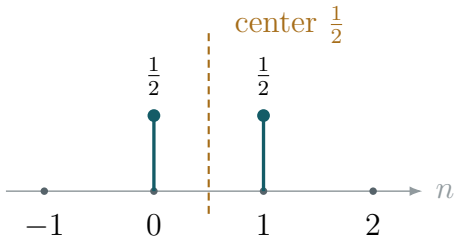


Figure 1.5: The moving-average impulse response. Two equal coefficients lie on opposite sides of a half-integer center. This symmetry will reappear in its phase response.

Two inputs make its smoothing effect explicit. If $x[n] = c$ is constant, then $y[n] = (c + c)/2 = c$. If $x[n] = (-1)^n$ alternates at every sample, then

$$y[n] = \tfrac{1}{2}\big((-1)^n + (-1)^{n-1}\big) = 0.$$

Thus the filter preserves a constant input but cancels a signal that changes sign at every sample. These are the discrete frequencies 0 and π, respectively. To determine its action at the remaining frequencies, we evaluate the response to a general complex exponential.

1.5 Pure frequencies and the frequency response

Write $\mathrm{j}^2 = -1$. Euler's identity $\mathrm{e}^{\mathrm{j}\theta} = \cos\theta + \mathrm{j}\sin\theta$ combines a cosine and sine into one convenient input. We extend a real linear filter to complex sequences by applying it separately to the real and imaginary parts.

For a pure frequency $x[n] = \mathrm{e}^{\mathrm{j}\omega n}$, convolution gives

$$\begin{aligned}(h * x)[n] &= \sum_{k\in\mathbb{Z}} h[k]\mathrm{e}^{\mathrm{j}\omega(n-k)} \\ &= \mathrm{e}^{\mathrm{j}\omega n}\sum_{k\in\mathbb{Z}} h[k]\mathrm{e}^{-\mathrm{j}\omega k} \\ &= H(\mathrm{e}^{\mathrm{j}\omega})\mathrm{e}^{\mathrm{j}\omega n}, \end{aligned} \tag{1.7}$$

where

$$H(\mathrm{e}^{\mathrm{j}\omega}) := \sum_{k\in\mathbb{Z}} h[k]\mathrm{e}^{-\mathrm{j}\omega k}. \tag{1.8}$$

The output is a scalar multiple of the input exponential. Thus, at a fixed frequency, the effect of convolution reduces to multiplication by the complex number $H(\mathrm{e}^{\mathrm{j}\omega})$. This is the analogue of an eigenvector calculation in linear algebra: the exponential is an eigenfunction of the convolution system, and the frequency response gives its eigenvalue.



Figure 1.6: An LTI filter does not create new frequencies from a single exponential. Its frequency response specifies the gain and phase at each frequency.

If $H(e^{j\omega}) = \rho e^{j\theta}$ with $\rho \geq 0$, then the output is $\rho e^{j(\omega n+\theta)}$. For a real filter, taking real parts yields

$$\cos(\omega n) \;\longmapsto\; \rho \cos(\omega n + \theta).$$

Thus $|H(e^{j\omega})|$ is the magnitude response, and an argument of $H(e^{j\omega})$ is the phase response when $H \neq 0$.

For an FIR filter, this calculation involves only a finite sum. It also holds when $\sum_k |h[k]| < \infty$, since $|e^{j\omega(n-k)}| = 1$ and the series is absolutely convergent.

1.6 Convolution becomes multiplication

For an absolutely summable sequence x, its *discrete-time Fourier transform* (DTFT) is

$$X(e^{j\omega}) \coloneqq \sum_{n\in\mathbb{Z}} x[n] e^{-j\omega n}.$$

The frequency response in (1.8) is precisely the DTFT of the impulse response. The DTFT is 2π-periodic, since replacing ω by $\omega + 2\pi$ multiplies the nth summand by $e^{-j2\pi n} = 1$. It therefore suffices to examine a single interval of length 2π, usually $[-\pi, \pi]$.

Proposition 1.1 (Convolution theorem)**.** If $\sum_n |x[n]| < \infty$ and $\sum_n |h[n]| < \infty$, then $y = h * x$ is absolutely summable and

$$Y(e^{j\omega}) = H(e^{j\omega}) X(e^{j\omega}). \tag{1.9}$$

Proof. The bound

$$\sum_n \sum_k |h[k]x[n-k]| = \sum_k |h[k]| \sum_m |x[m]| < \infty$$

shows both that y is absolutely summable and that we can reorder the double series below. Substitute $m = n - k$:

$$\begin{aligned} Y(\mathrm{e}^{\mathrm{j}\omega}) &= \sum_n \left(\sum_k h[k]x[n-k] \right) \mathrm{e}^{-\mathrm{j}\omega n} \\ &= \sum_k h[k] \sum_m x[m] \mathrm{e}^{-\mathrm{j}\omega(m+k)} \\ &= \left(\sum_k h[k] \mathrm{e}^{-\mathrm{j}\omega k} \right) \left(\sum_m x[m] \mathrm{e}^{-\mathrm{j}\omega m} \right). \end{aligned}$$

The two factors are H and X. □

The z-transform. Define

$$X(z) \coloneqq \sum_{n \in \mathbb{Z}} x[n] z^{-n}, \qquad z \in \mathbb{C} \setminus \{0\}, \tag{1.10}$$

The z-transform is defined where the series converges. On the unit circle, whenever this substitution is valid,

$$X(\mathrm{e}^{\mathrm{j}\omega}) = X(z)\big|_{z=\mathrm{e}^{\mathrm{j}\omega}}.$$

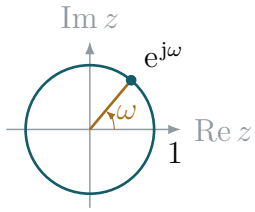


Figure 1.7: The frequency response is the z-transform of the filter evaluated on the unit circle.

1.7 The moving average in the frequency domain

For $h[n] = \frac{1}{2}(\delta[n] + \delta[n-1])$, the z-transform is

$$H(z) = \tfrac{1}{2}(1 + z^{-1}).$$

Evaluate on the unit circle and factor around the midpoint of the two powers:

$$\begin{aligned} H(\mathrm{e}^{\mathrm{j}\omega}) &= \tfrac{1}{2} + \tfrac{1}{2}\mathrm{e}^{-\mathrm{j}\omega} \\ &= \mathrm{e}^{-\mathrm{j}\omega/2} \frac{\mathrm{e}^{\mathrm{j}\omega/2} + \mathrm{e}^{-\mathrm{j}\omega/2}}{2} \\ &= \underbrace{\mathrm{e}^{-\mathrm{j}\omega/2}}_{\text{phase factor}} \underbrace{\cos(\omega/2)}_{\text{real amplitude}}. \end{aligned} \tag{1.11}$$

The last step is Euler's cosine identity. The exponential factor has magnitude one, so

$$|H(\mathrm{e}^{\mathrm{j}\omega})| = |\cos(\omega/2)|.$$

On $-\pi < \omega < \pi$, the cosine is positive. A continuous phase choice on this interval is therefore $\varphi(\omega) = -\omega/2$. At $\omega = \pm\pi$ the response vanishes, and its phase is undefined.



Figure 1.8: The moving average preserves zero frequency and suppresses frequencies near $\pm\pi$. Its phase has constant slope $-1/2$. Open endpoints on the phase plot indicate zeros of the response.

The low-pass interpretation is now quantitative. At $\omega = 0$, the gain is one. At $\omega = \pi$, it is zero. At an intermediate frequency such as $\omega = \pi/2$, it is $\cos(\pi/4) = 1/\sqrt{2}$. These agree with our constant-input and alternating-input calculations.

The phase slope $-1/2$ is determined by the center of the two filter coefficients, which is $L = 1/2$. After factoring out the exponential corresponding to this center, the remaining terms have equal and opposite offsets. They therefore combine into a cosine. The same calculation applies to every symmetric FIR filter.

1.8 Linear phase and a center of symmetry

For a real filter, we use the linear-phase factorization

$$H(\mathrm{e}^{\mathrm{j}\omega}) = \mathrm{e}^{\mathrm{j}(-L\omega+\beta)} H_{\mathrm{ampl}}(\omega), \qquad H_{\mathrm{ampl}}(\omega) \in \mathbb{R}. \tag{1.12}$$

The phase factor is affine in frequency: its slope is $-L$ and its offset is β. In DSP this is called *linear phase*, including the constant offset.

> **Real amplitude is not necessarily magnitude**
>
> The real function H_{ampl} may change sign. The magnitude response is $|H_{\mathrm{ampl}}|$. Where $H_{\mathrm{ampl}} < 0$, the actual phase has an additional π modulo 2π; at zeros it is undefined. Thus (1.12) does not assert that a principal-value phase plot is one unbroken line.

Consider an FIR filter whose first and last nonzero coefficients occur at 0 and $N-1$. Its *length* is N, and its center is

$$L \coloneqq \frac{N-1}{2}.$$

Length includes any zero coefficients between the endpoints. For example, a five-sample antisymmetric filter can have a zero middle coefficient and still have length five.

Symmetry and antisymmetry about L mean, respectively,

$$h[n] = h[2L-n], \qquad h[n] = -h[2L-n]. \tag{1.13}$$

The index $2L-n$ is the reflection of n about L. For a pair of locations $L-r$ and $L+r$, factor out the center exponential:

$$\begin{aligned} &h[L-r]\mathrm{e}^{-\mathrm{j}(L-r)\omega} + h[L+r]\mathrm{e}^{-\mathrm{j}(L+r)\omega} \\ &\qquad = \mathrm{e}^{-\mathrm{j}L\omega}\big(h[L-r]\mathrm{e}^{\mathrm{j}r\omega} + h[L+r]\mathrm{e}^{-\mathrm{j}r\omega}\big). \end{aligned}$$

If the coefficients are equal, the expression in parentheses is $2h[L-r]\cos(r\omega)$. If they have opposite signs, it is $2\mathrm{j}h[L-r]\sin(r\omega)$. There are therefore two independent distinctions: symmetry versus antisymmetry, and odd versus even length. Together, they give the four linear-phase FIR types:

	N odd, $L \in \mathbb{Z}$	N even, $L - \frac{1}{2} \in \mathbb{Z}$
symmetric	Type I	Type II
antisymmetric	Type III	Type IV

1.9 Type I: odd length, symmetric

Here $N = 2L+1$ with L an integer. There is a coefficient at the center, and the other coefficients occur in L reflected pairs.

$$\begin{aligned} H(\mathrm{e}^{\mathrm{j}\omega}) &= h[L]\mathrm{e}^{-\mathrm{j}L\omega} + \sum_{r=1}^{L} h[L-r]\big(\mathrm{e}^{-\mathrm{j}(L-r)\omega} + \mathrm{e}^{-\mathrm{j}(L+r)\omega}\big) \\ &= \mathrm{e}^{-\mathrm{j}L\omega}\left(h[L] + 2\sum_{r=1}^{L} h[L-r]\cos(r\omega)\right). \end{aligned} \tag{1.14}$$

Thus H_{ampl} is a sum of integer-frequency cosines, and $\beta = 0$.

Example. Take

$$(h[0], \ldots, h[4]) = (\tfrac{1}{4}, \tfrac{1}{2}, 1, \tfrac{1}{2}, \tfrac{1}{4}), \qquad L = 2.$$

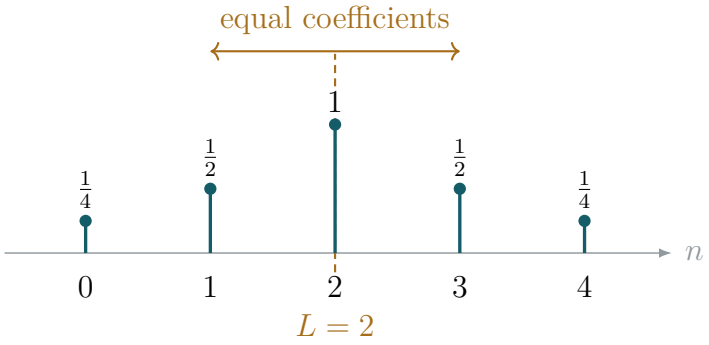


Figure 1.9: An integer center leaves one unpaired coefficient. The two neighboring pairs contribute cosines at offsets one and two.

The frequency response is

$$\begin{aligned} H(\mathrm{e}^{\mathrm{j}\omega}) &= \tfrac{1}{4} + \tfrac{1}{2}\mathrm{e}^{-\mathrm{j}\omega} + \mathrm{e}^{-2\mathrm{j}\omega} + \tfrac{1}{2}\mathrm{e}^{-3\mathrm{j}\omega} + \tfrac{1}{4}\mathrm{e}^{-4\mathrm{j}\omega} \\ &= \mathrm{e}^{-2\mathrm{j}\omega}\left[1 + \tfrac{1}{2}(\mathrm{e}^{\mathrm{j}\omega} + \mathrm{e}^{-\mathrm{j}\omega}) + \tfrac{1}{4}(\mathrm{e}^{2\mathrm{j}\omega} + \mathrm{e}^{-2\mathrm{j}\omega})\right] \\ &= \mathrm{e}^{-2\mathrm{j}\omega}\left[1 + \cos\omega + \tfrac{1}{2}\cos(2\omega)\right]. \end{aligned}$$

Express the cosine coefficients in terms of h.

In the notation $H_{\mathrm{ampl}}(\omega) = \sum_{r=0}^{L} a[r]\cos(r\omega)$, equation (1.14) gives

$$a[0] = h[L], \qquad a[r] = 2h[L-r] \quad (1 \le r \le L).$$

The center is counted once; every reflected pair is counted twice. That distinction explains the different formula for $a[0]$.

1.10 Type II: even length, symmetric

For an even-length symmetric filter, $L - \frac{1}{2}$ is an integer. The pairing argument is the same as for Type I, but there is no sample at the center. The offsets from the center are $r + \frac{1}{2}$, where $r = 0, \ldots, L - \frac{1}{2}$.

Pairing the two equal coefficients at each offset gives

$$\begin{aligned} H(\mathrm{e}^{\mathrm{j}\omega}) &= \sum_{r=0}^{L-1/2} h[L-r-\tfrac{1}{2}]\left(\mathrm{e}^{-\mathrm{j}(L-r-1/2)\omega} + \mathrm{e}^{-\mathrm{j}(L+r+1/2)\omega}\right) \\ &= \mathrm{e}^{-\mathrm{j}L\omega} \sum_{r=0}^{L-1/2} 2h[L-r-\tfrac{1}{2}]\cos\big((r+\tfrac{1}{2})\omega\big). \end{aligned} \tag{1.15}$$

All indices of h are integers, even though L itself is a half-integer. Again $\beta = 0$, but the cosine arguments now involve half-integer offsets.

Example. Take $(h[0], \ldots, h[3]) = (\frac{1}{2}, 1, 1, \frac{1}{2})$, so $L = \frac{3}{2}$.

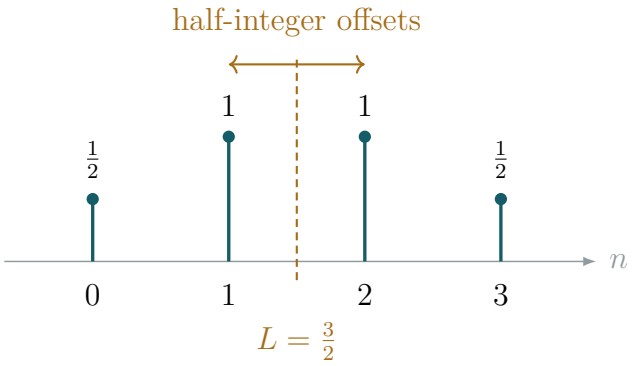


Figure 1.10: The closest samples are half a sample from the center, not one sample. This changes integer cosines into half-integer cosines.

The four terms can be grouped as

$$\begin{aligned} H(\mathrm{e}^{\mathrm{j}\omega}) &= \tfrac{1}{2} + \mathrm{e}^{-\mathrm{j}\omega} + \mathrm{e}^{-2\mathrm{j}\omega} + \tfrac{1}{2}\mathrm{e}^{-3\mathrm{j}\omega} \\ &= \mathrm{e}^{-3\mathrm{j}\omega/2}\left[(\mathrm{e}^{\mathrm{j}\omega/2} + \mathrm{e}^{-\mathrm{j}\omega/2}) + \tfrac{1}{2}(\mathrm{e}^{3\mathrm{j}\omega/2} + \mathrm{e}^{-3\mathrm{j}\omega/2})\right] \\ &= \mathrm{e}^{-3\mathrm{j}\omega/2}\left[2\cos(\omega/2) + \cos(3\omega/2)\right]. \end{aligned}$$

Thus the coefficients in $H_{\text{ampl}}(\omega) = \sum_{r=0}^{L-1/2} b[r]\cos((r+\frac{1}{2})\omega)$ are

$$b[r] = 2h[L - r - \tfrac{1}{2}], \qquad 0 \le r \le L - \tfrac{1}{2}.$$

The moving average is the shortest Type II example: $L = \frac{1}{2}$ and $b[0] = 1$.

1.11 Antisymmetry: the center sample matters

We have seen how equal reflected coefficients produce cosine terms. We now keep the same pairing of locations and change the relation between the coefficients. For an antisymmetric filter, reflection changes the sign:

$$h[n] = -h[2L - n].$$

When L is an integer, substituting $n = L$ gives

$$h[L] = -h[L] \quad \Longrightarrow \quad \boxed{h[L] = 0.}$$

The reflection rule applies to every coefficient, including a coefficient located at the center. Consequently, an odd-length antisymmetric filter must have a zero center coefficient. Opposite values in the remaining pairs do not compensate for a nonzero value at the center.

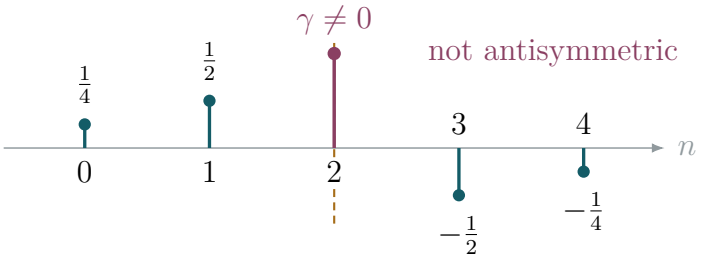


Figure 1.11: The nonzero center coefficient breaks antisymmetry, even though the outer pairs have opposite values. The symbol γ denotes the nonzero center height.

1.12 Type III: odd length, antisymmetric

For integer L, the center term vanishes and the reflected coefficients have opposite signs. Pairing them yields

$$\begin{aligned} H(\mathrm{e}^{\mathrm{j}\omega}) &= \sum_{r=1}^{L} h[L-r]\big(\mathrm{e}^{-\mathrm{j}(L-r)\omega} - \mathrm{e}^{-\mathrm{j}(L+r)\omega}\big) \\ &= \mathrm{e}^{-\mathrm{j}L\omega} \sum_{r=1}^{L} h[L-r]\big(\mathrm{e}^{\mathrm{j}r\omega} - \mathrm{e}^{-\mathrm{j}r\omega}\big) \\ &= \mathrm{e}^{\mathrm{j}(-L\omega+\pi/2)} \sum_{r=1}^{L} 2h[L-r]\sin(r\omega). \end{aligned} \tag{1.16}$$

We used $\mathrm{e}^{\mathrm{j}\theta} - \mathrm{e}^{-\mathrm{j}\theta} = 2\,\mathrm{j}\sin\theta$ and $\mathrm{j} = \mathrm{e}^{\mathrm{j}\pi/2}$. Thus antisymmetry produces a sine amplitude and a constant phase offset $\beta = \pi/2$.

Type III example. Take

$$(h[0], \ldots, h[4]) = (\tfrac{1}{2}, 1, 0, -1, -\tfrac{1}{2}), \qquad L = 2.$$

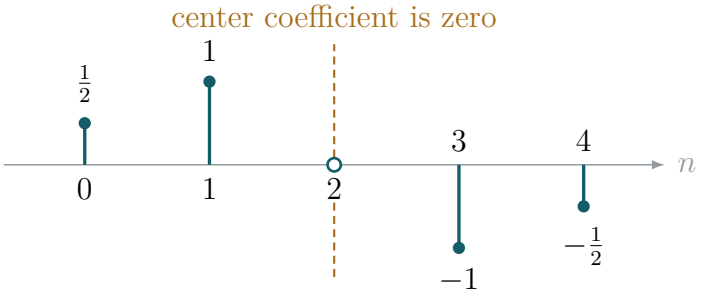


Figure 1.12: Opposite coefficients at integer offsets from the center. The nearest pair gives $2\sin\omega$; the outer pair gives $\sin(2\omega)$.

Expanding and then pairing gives

$$\begin{aligned} H(\mathrm{e}^{\mathrm{j}\omega}) &= \tfrac{1}{2} + \mathrm{e}^{-\mathrm{j}\omega} - \mathrm{e}^{-3\mathrm{j}\omega} - \tfrac{1}{2}\mathrm{e}^{-4\mathrm{j}\omega} \\ &= \mathrm{e}^{-2\mathrm{j}\omega}\left[(\mathrm{e}^{\mathrm{j}\omega} - \mathrm{e}^{-\mathrm{j}\omega}) + \tfrac{1}{2}(\mathrm{e}^{2\mathrm{j}\omega} - \mathrm{e}^{-2\mathrm{j}\omega})\right] \\ &= \mathrm{e}^{-2\mathrm{j}\omega}\,\mathrm{j}\,[2\sin\omega + \sin(2\omega)] \\ &= \mathrm{e}^{\mathrm{j}(-2\omega+\pi/2)}\,[2\sin\omega + \sin(2\omega)]\,. \end{aligned}$$

Writing the general amplitude as $\sum_{r=0}^{L} c[r]\sin(r\omega)$, we may take

$$c[0] = 0, \qquad c[r] = 2h[L-r] \quad (1 \le r \le L).$$

The choice of $c[0]$ has no effect because $\sin(0\omega) = 0$.

Compared with the symmetric case, antisymmetry introduces a factor j, which contributes the constant phase offset $\pi/2$. The slope of the phase factor remains $-L$. After this phase factor is removed, the real amplitude is a sine sum rather than a cosine sum. Its possible sign changes must still be accounted for when interpreting a plotted phase response.

1.13 Type IV: even length, antisymmetric

Here $L - \frac{1}{2}$ is an integer. There is no center sample to force to zero. Each pair is at offsets $\pm(r + \frac{1}{2})$, and its coefficients have opposite signs:

$$\begin{aligned} H(\mathrm{e}^{\mathrm{j}\omega}) &= \sum_{r=0}^{L-1/2} h[L-r-\tfrac{1}{2}]\left(\mathrm{e}^{-\mathrm{j}(L-r-1/2)\omega} - \mathrm{e}^{-\mathrm{j}(L+r+1/2)\omega}\right) \\ &= \mathrm{e}^{-\mathrm{j}L\omega}\sum_{r=0}^{L-1/2} h[L-r-\tfrac{1}{2}]\left(\mathrm{e}^{\mathrm{j}(r+1/2)\omega} - \mathrm{e}^{-\mathrm{j}(r+1/2)\omega}\right) \\ &= \mathrm{e}^{\mathrm{j}(-L\omega+\pi/2)}\sum_{r=0}^{L-1/2} 2h[L-r-\tfrac{1}{2}]\sin\big((r+\tfrac{1}{2})\omega\big). \end{aligned} \tag{1.17}$$

Thus the amplitude uses half-integer sine arguments and $\beta = \pi/2$.

Example. For $(h[0], \ldots, h[3]) = (\frac{1}{2}, 1, -1, -\frac{1}{2})$, the center is $L = \frac{3}{2}$.

Factoring the center exponential gives

$$\begin{aligned} H(\mathrm{e}^{\mathrm{j}\omega}) &= \tfrac{1}{2} + \mathrm{e}^{-\mathrm{j}\omega} - \mathrm{e}^{-2\mathrm{j}\omega} - \tfrac{1}{2}\mathrm{e}^{-3\mathrm{j}\omega} \\ &= \mathrm{e}^{-3\mathrm{j}\omega/2}\left[(\mathrm{e}^{\mathrm{j}\omega/2} - \mathrm{e}^{-\mathrm{j}\omega/2}) + \tfrac{1}{2}(\mathrm{e}^{3\mathrm{j}\omega/2} - \mathrm{e}^{-3\mathrm{j}\omega/2})\right] \\ &= \mathrm{e}^{\mathrm{j}(-3\omega/2+\pi/2)}\,[2\sin(\omega/2) + \sin(3\omega/2)]\,. \end{aligned}$$

In the general amplitude $\sum_{r=0}^{L-1/2} d[r]\sin((r+\frac{1}{2})\omega)$,

$$d[r] = 2h[L-r-\tfrac{1}{2}], \qquad 0 \le r \le L - \tfrac{1}{2}.$$

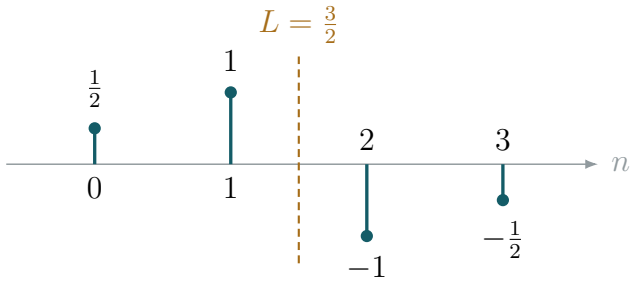


Figure 1.13: Opposite coefficients at half-integer offsets. There is no unpaired sample at the center.

1.14 Why these four cases give the characterization

The pair calculations prove that each type has linear phase. Conversely, suppose a real FIR filter has nonzero endpoints at 0 and $N-1$ and admits

$$H(\mathrm{e}^{\mathrm{j}\omega}) = \mathrm{e}^{\mathrm{j}(-L\omega+\beta)} H_{\mathrm{ampl}}(\omega)$$

with real H_{ampl}. We will recover both the center and the symmetry.

Conjugate this factorization and eliminate the real amplitude:

$$H(\mathrm{e}^{\mathrm{j}\omega}) = \mathrm{e}^{-2\mathrm{j}L\omega+2\mathrm{j}\beta} H(\mathrm{e}^{\mathrm{j}\omega})^*. \tag{1.18}$$

Since H is 2π-periodic and not identically zero, replacing ω by $\omega + 2\pi$ forces $\mathrm{e}^{-4\pi\mathrm{j}L} = 1$. Thus $2L$ is an integer. Extend h by zero outside its support and reindex the finite sum:

$$\begin{aligned} H(\mathrm{e}^{\mathrm{j}\omega}) &= \mathrm{e}^{2\mathrm{j}\beta} \sum_{n\in\mathbb{Z}} h[n] \mathrm{e}^{-\mathrm{j}(2L-n)\omega} \\ &= \mathrm{e}^{2\mathrm{j}\beta} \sum_{m\in\mathbb{Z}} h[2L-m] \mathrm{e}^{-\mathrm{j}m\omega}. \end{aligned}$$

To compare this with $\sum_m h[m]\mathrm{e}^{-\mathrm{j}m\omega}$, multiply by $\mathrm{e}^{\mathrm{j}r\omega}$ and integrate over $[-\pi, \pi]$. For integer q,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{e}^{\mathrm{j}q\omega}\,\mathrm{d}\omega = \begin{cases} 1, & q = 0, \\ \dfrac{\mathrm{e}^{\mathrm{j}q\pi} - \mathrm{e}^{-\mathrm{j}q\pi}}{2\pi\,\mathrm{j}q} = 0, & q \neq 0. \end{cases}$$

Only the term with $m = r$ survives. Therefore

$$h[r] = \mathrm{e}^{2\mathrm{j}\beta} h[2L-r] \qquad (r \in \mathbb{Z}). \tag{1.19}$$

Since $h[0] \neq 0$, we have $h[2L] \neq 0$, hence $2L \leq N-1$. Since $h[N-1] \neq 0$, we have $h[2L-(N-1)] \neq 0$, hence $2L \geq N-1$. Consequently $L = (N-1)/2$.

The two nonzero endpoint values in (1.19) are real. Thus $\mathrm{e}^{2\mathrm{j}\beta}$ is real and has modulus one: it is 1 or -1. These give symmetry and antisymmetry, respectively.

We may choose $\beta = 0$ or $\pi/2$, absorbing any remaining minus sign into H_{ampl}. Finally, the length is odd or even, giving exactly Types I–IV. The assumptions of a *real, nonzero FIR filter* are essential to this statement.

1.15 The four types at a glance

In every case,

$$H(e^{j\omega}) = e^{j(-L\omega+\beta)} H_{\text{ampl}}(\omega), \qquad L = \frac{N-1}{2}.$$

The center controls the phase slope. The sign under reflection controls whether the amplitude is a cosine sum or a sine sum. The parity of the length controls whether the offsets are integers or half-integers.

TYPE I
odd length, symmetric

$$H_{\text{ampl}} = \sum_{r=0}^{L} a[r] \cos(r\omega)$$

$L \in \mathbb{Z}, \quad \beta = 0$

TYPE II
even length, symmetric

$$H_{\text{ampl}} = \sum_{r=0}^{L-1/2} b[r] \cos((r+\tfrac{1}{2})\omega)$$

$L - \frac{1}{2} \in \mathbb{Z}, \quad \beta = 0$

TYPE III
odd length, antisymmetric

$$H_{\text{ampl}} = \sum_{r=1}^{L} c[r] \sin(r\omega)$$

$L \in \mathbb{Z}, \quad \beta = \frac{\pi}{2}, \quad h[L] = 0$

TYPE IV
even length, antisymmetric

$$H_{\text{ampl}} = \sum_{r=0}^{L-1/2} d[r] \sin((r+\tfrac{1}{2})\omega)$$

$L - \frac{1}{2} \in \mathbb{Z}, \quad \beta = \frac{\pi}{2}$

Figure 1.14: Two choices give four types. The formulas were obtained by pairing coefficients at equal distances from the center, not by memorizing four unrelated identities.

Many wavelet filters have linear phase. The symmetry and antisymmetry conditions above will therefore be useful when we study wavelet filter banks.

The amplitudes in the examples were not normalized to have unit gain at zero frequency. For instance, the Type I example has $H(1) = 1+1+\frac{1}{2} = \frac{5}{2}$. Multiplying all its coefficients by $2/5$ would normalize this gain without changing its type or its center. Symmetry and gain normalization are separate properties.

Likewise, the type alone does not say whether a filter is low-pass, high-pass, or band-pass. The coefficients determine the amplitude response. But the type can force zeros at particular frequencies, which may rule out a proposed use. The final question gives an example.

1.16 Can a high-pass filter be Type II?

Consider a high-pass specification whose passband includes the Nyquist frequency $\omega = \pi$, so that the required gain there is nonzero. A Type II filter cannot satisfy this specification.

Indeed, every term in its real amplitude vanishes at $\omega = \pi$:

$$\cos\big((r + \tfrac{1}{2})\pi\big) = \cos(r\pi + \pi/2) = 0 \qquad (r \in \mathbb{Z}).$$

Substitute this in (1.15):

$$\begin{aligned} H(\mathrm{e}^{\mathrm{j}\pi}) &= \mathrm{e}^{-\mathrm{j}L\pi} \sum_{r=0}^{L-1/2} b[r] \cos\big((r + \tfrac{1}{2})\pi\big) \\ &= 0. \end{aligned}$$

Thus a zero at π is forced by the combination of even length and symmetry, independently of the values of $b[r]$.

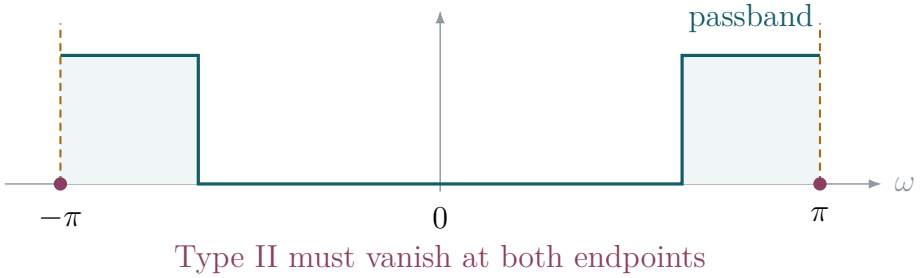


Figure 1.15: A desired high-pass magnitude with nonzero endpoint gain. The filled dots mark the zeros that every Type II filter must have. The rectangular response is a specification sketch, not a claim of an exact FIR realization.

What to remember

Linearity and time invariance turn an input into a superposition of shifted impulse responses: $y = h * x$. Complex exponentials turn that convolution into a scalar multiplication. For real FIR filters, pairing coefficients about their center exposes linear phase: equal pairs give cosines, and opposite pairs give sines.

These examples illustrate how time-domain structure constrains frequency-domain behavior. Symmetry is a finite collection of relations among the filter coefficients, but it determines the phase at every frequency. In wavelet filter design, we will combine such relations with perfect-reconstruction and orthogonality conditions. The calculations above distinguish the effects of symmetry, normalization, and reconstruction, so that we can track each condition separately.